Controllers for Underwater Vehicle

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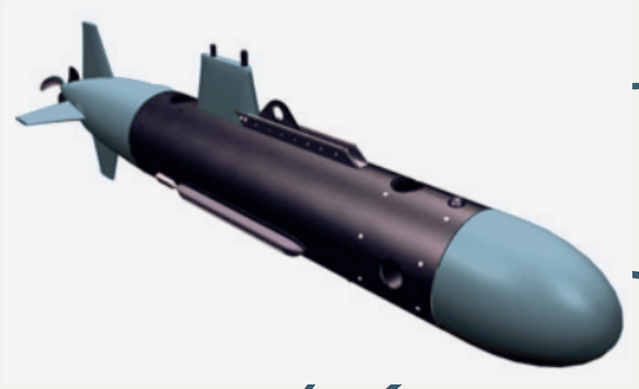
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Introduction







Underwater Robotics

These systems are now used in a broad spectrum of applications ranging from environmental monitoring, to underwater infrastructure maintenance, to scientific and military missions.

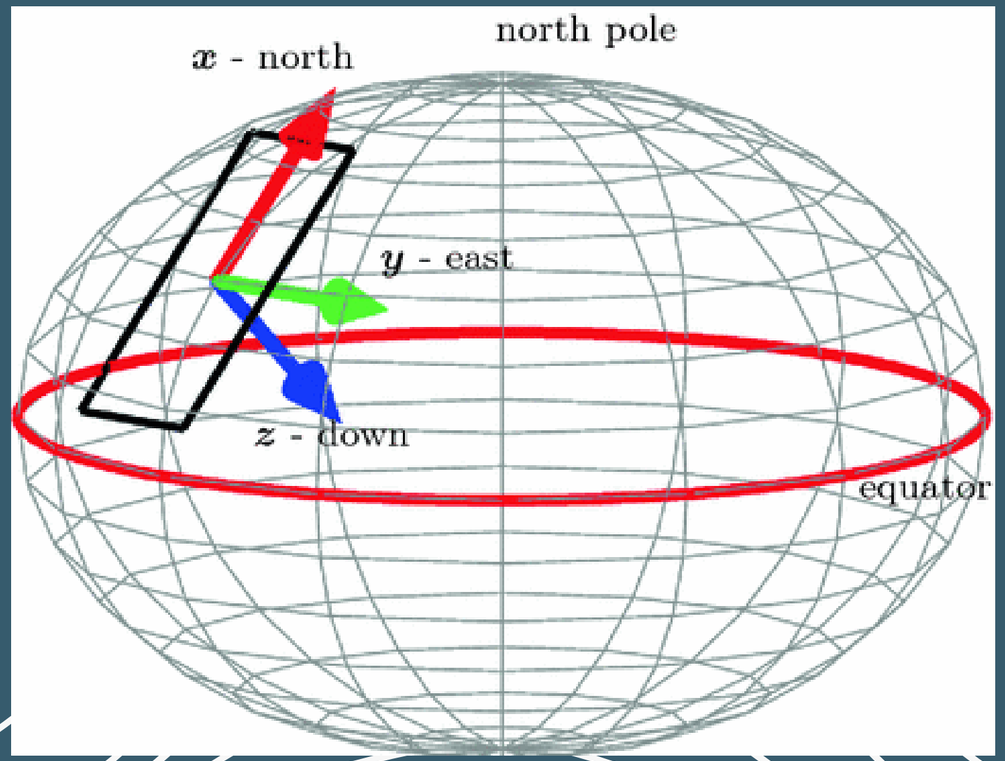
Precisely controlling the motion of an AUV poses a significant technological and theoretical challenge due to the complexity of the operational environment

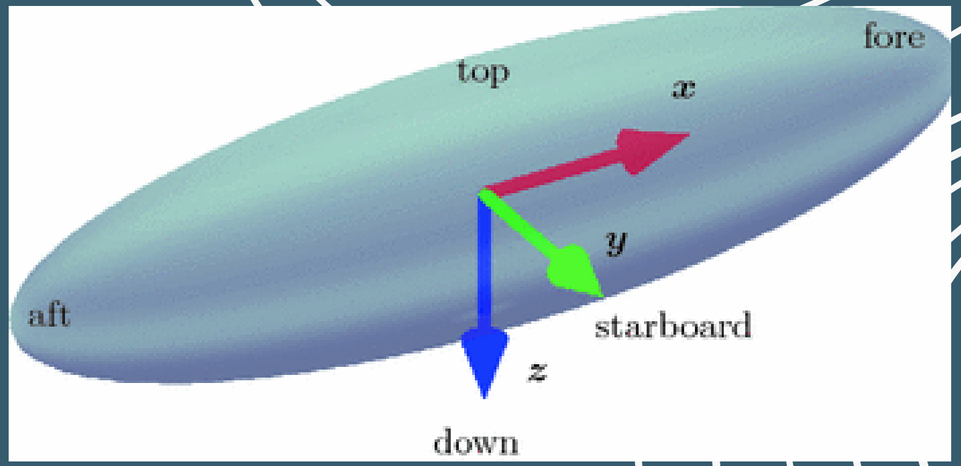
Frames

Reference Frame

Inertial (Earth-Fixed) Frame

Body-Fixed Frame





Dynamic equations

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu})\boldsymbol{\nu} + \boldsymbol{g}(\mathbf{R}_B^I) = \boldsymbol{\tau} - \boldsymbol{\tau}_{\text{current}}$$

- Added Mass
- Damping Effect
- Current Effect
- Restoring Forces

$$\boldsymbol{M}_v^\star(\boldsymbol{R}_B^I)\ddot{\boldsymbol{\eta}} + \boldsymbol{C}_v^\star(\boldsymbol{R}_B^I, \dot{\boldsymbol{\eta}})\dot{\boldsymbol{\eta}} + \boldsymbol{D}_{RB}^\star(\boldsymbol{R}_B^I, \dot{\boldsymbol{\eta}})\dot{\boldsymbol{\eta}} + \boldsymbol{g}_{RB}^\star(\boldsymbol{R}_B^I) = \boldsymbol{\tau}_v^\star,$$

Dynamics

Added Mass

As the vehicle accelerates, the surrounding fluid contributes to its inertia. This effect is captured by the added mass matrix, which depends on the vehicle geometry that affects the inertia matrix.

Dumping Effect

The viscosity of the fluid also causes the presence of dissipative drag and lift forces on the body.

Restoring Forces

When a rigid body is immersed in a fluid and subject to gravity, two additional forces must be taken into account: the gravitational force and the buoyancy.

Ocean Currents

Ocean current represent a slowly time-varying or constant disturbance

Controllers

Model-Based Controller

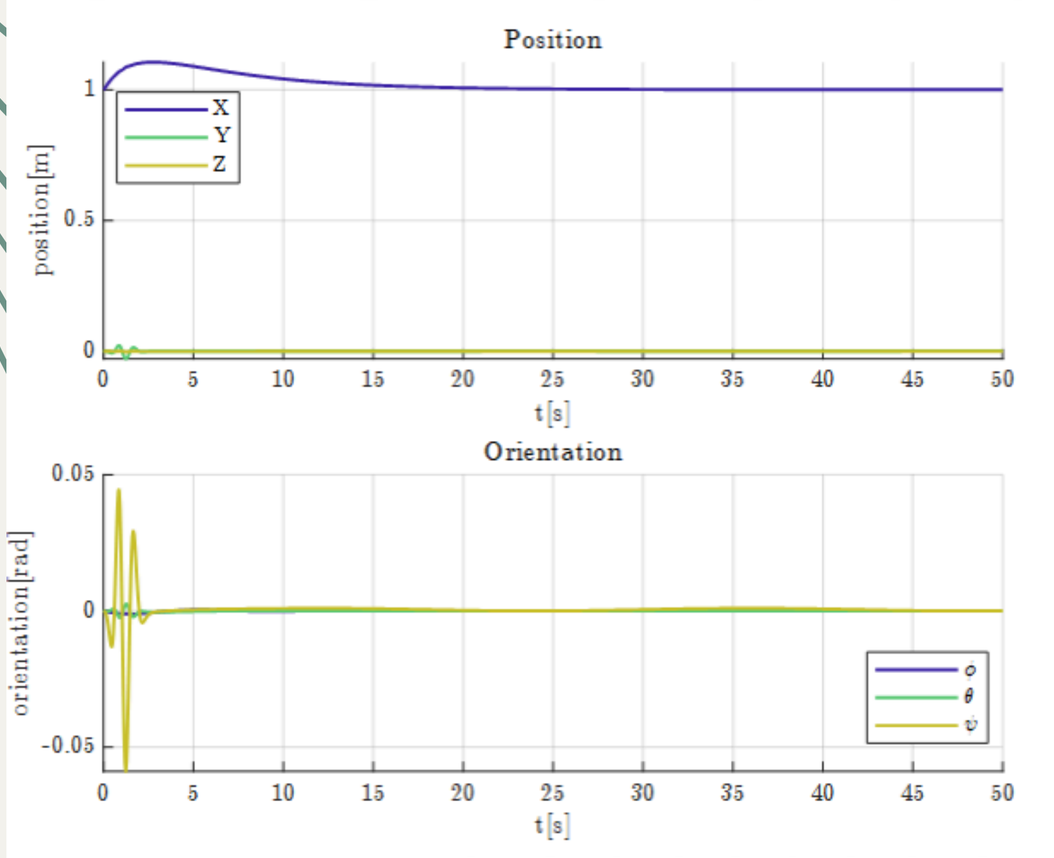
The first controller adopts a model-based law expressed in the earth-fixed frame.
This approach allows accurate compensation of vehicle dynamics by exploiting a comprehensive mathematical model

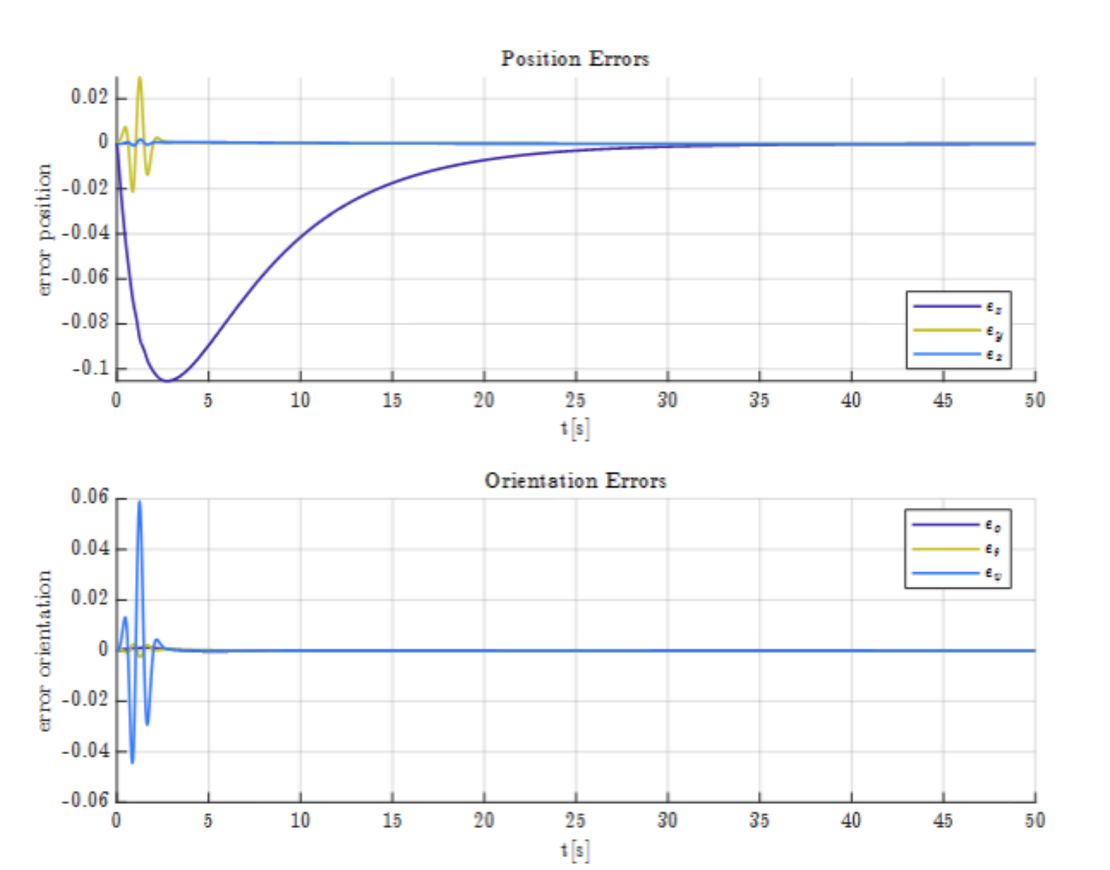
The control uses a PID term with feedforward compensation of forces and moments generated by inertia, Coriolis effects, damping, and buoyancy forces.

$$\tau = M\ddot{p}_r + C(v)\dot{p}_r + D(v)\dot{p}_r + g_{RB} + \Lambda s$$

Model-Based Controller

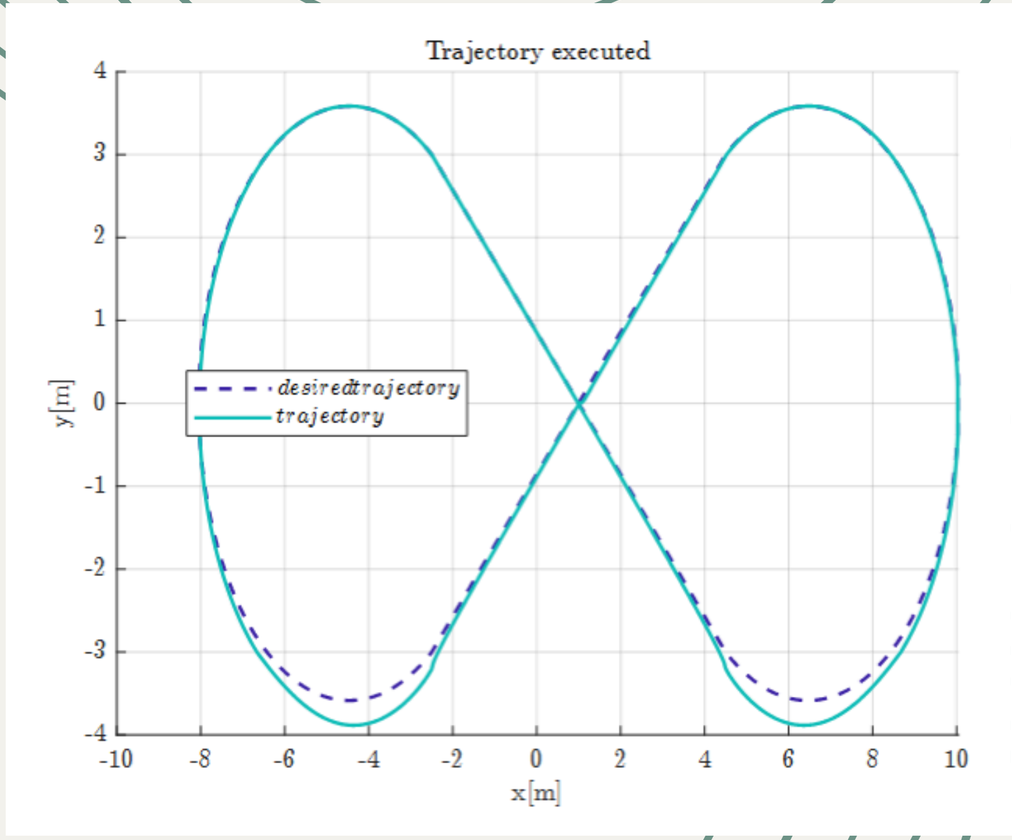
Regulation

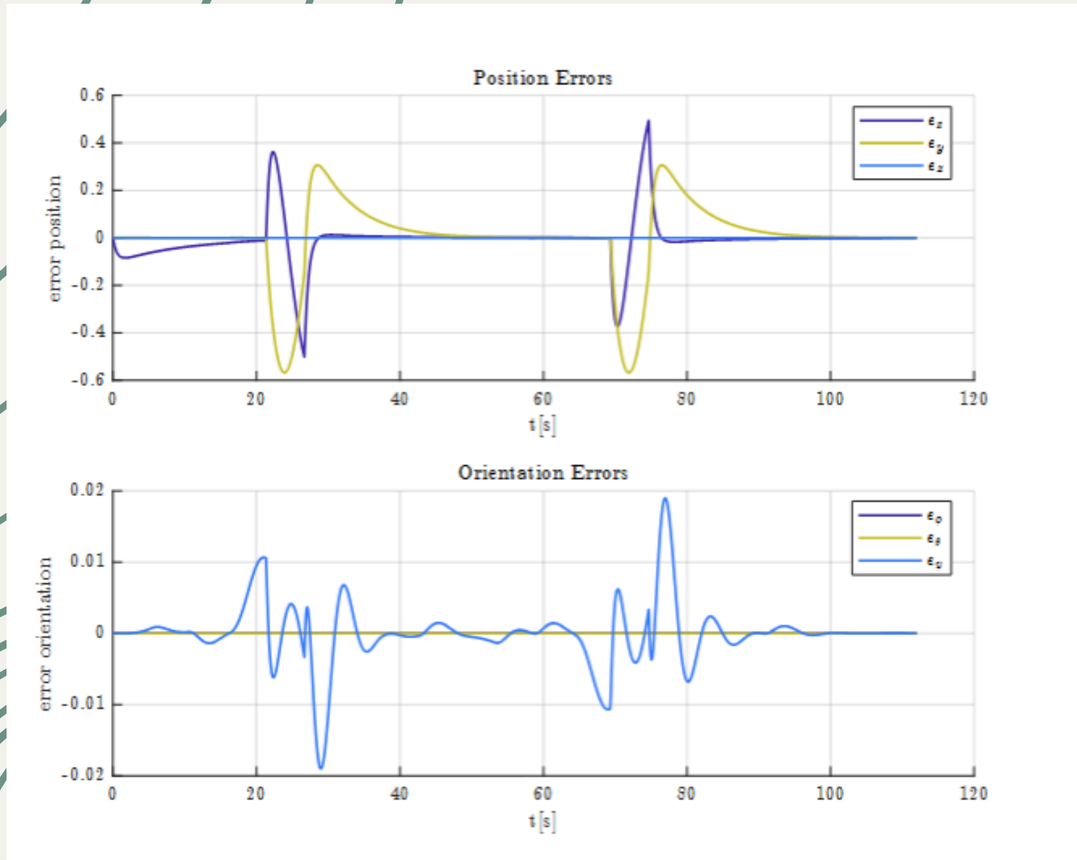




Model-Based Controller

Tracking





Controllers

Adaptive Control with Hydrodynamic Compensation

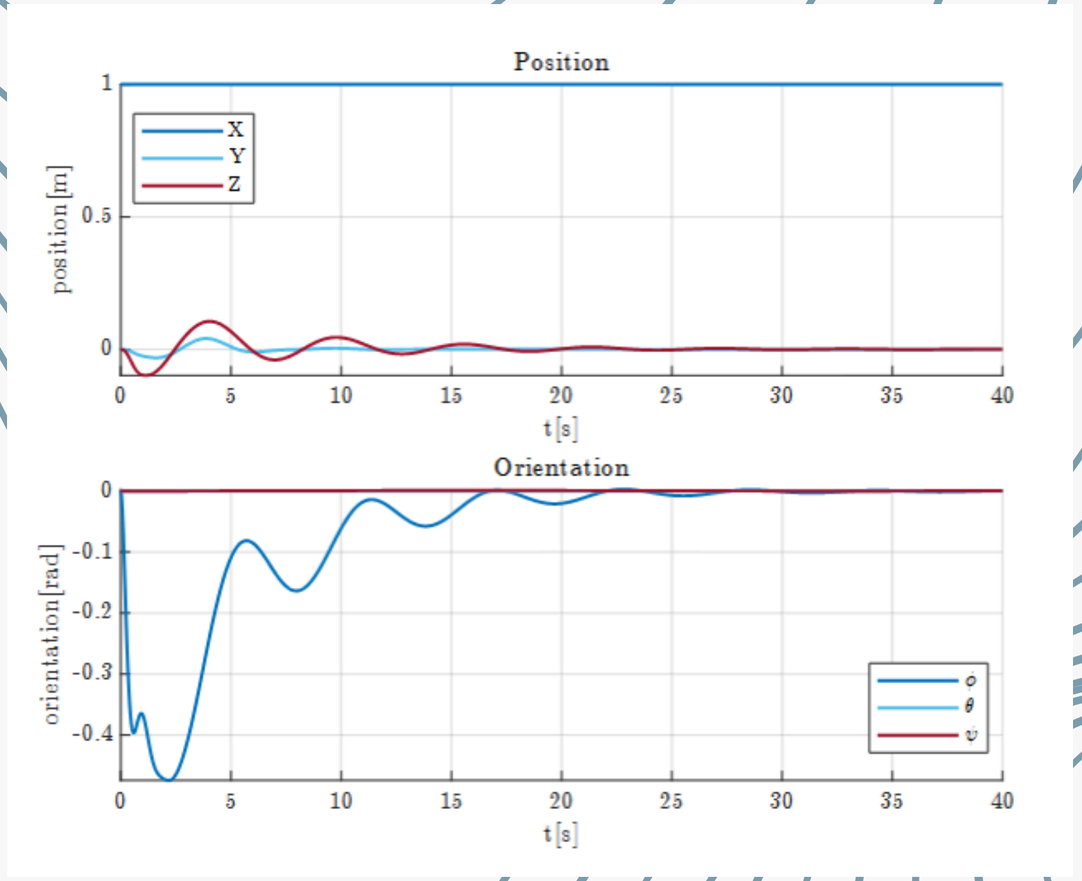
The second control is based on a hybrid formulation To effectively compensate for persistent hydrodynamic effects, such a buoyancy forces and the influence of sea current, an adaptive control law based on a linear formulation of the dynamic parameters was adopted.

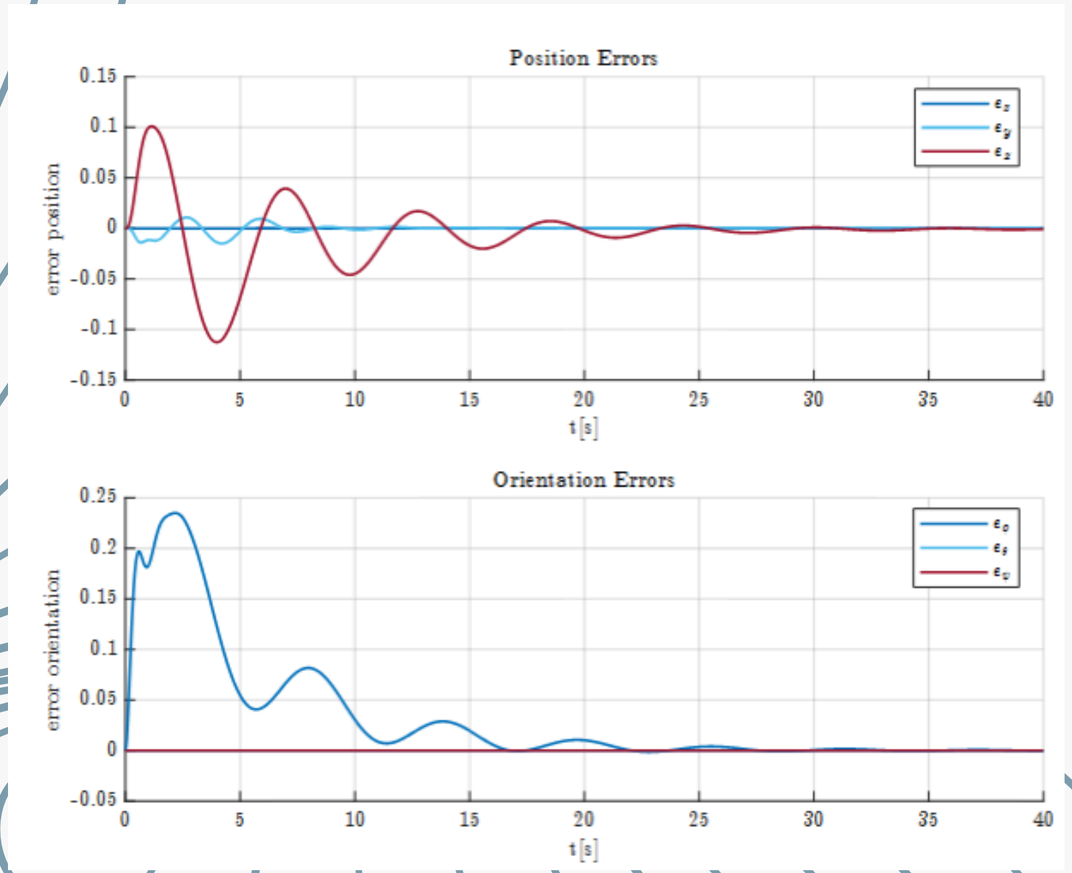
$$\boldsymbol{\tau} = \mathbf{K}_D\mathbf{s} + \mathbf{K}\tilde{\mathbf{y}} + \boldsymbol{\Phi}_P\boldsymbol{\lambda}$$

$$\dot{\boldsymbol{\lambda}} = -\mathbf{K}_\theta^{-1}\boldsymbol{\Phi}_P^T\mathbf{s} \qquad \mathbf{s} = \tilde{\boldsymbol{\nu}} + \boldsymbol{\Lambda}\tilde{\mathbf{y}}$$

Adaptive Controller

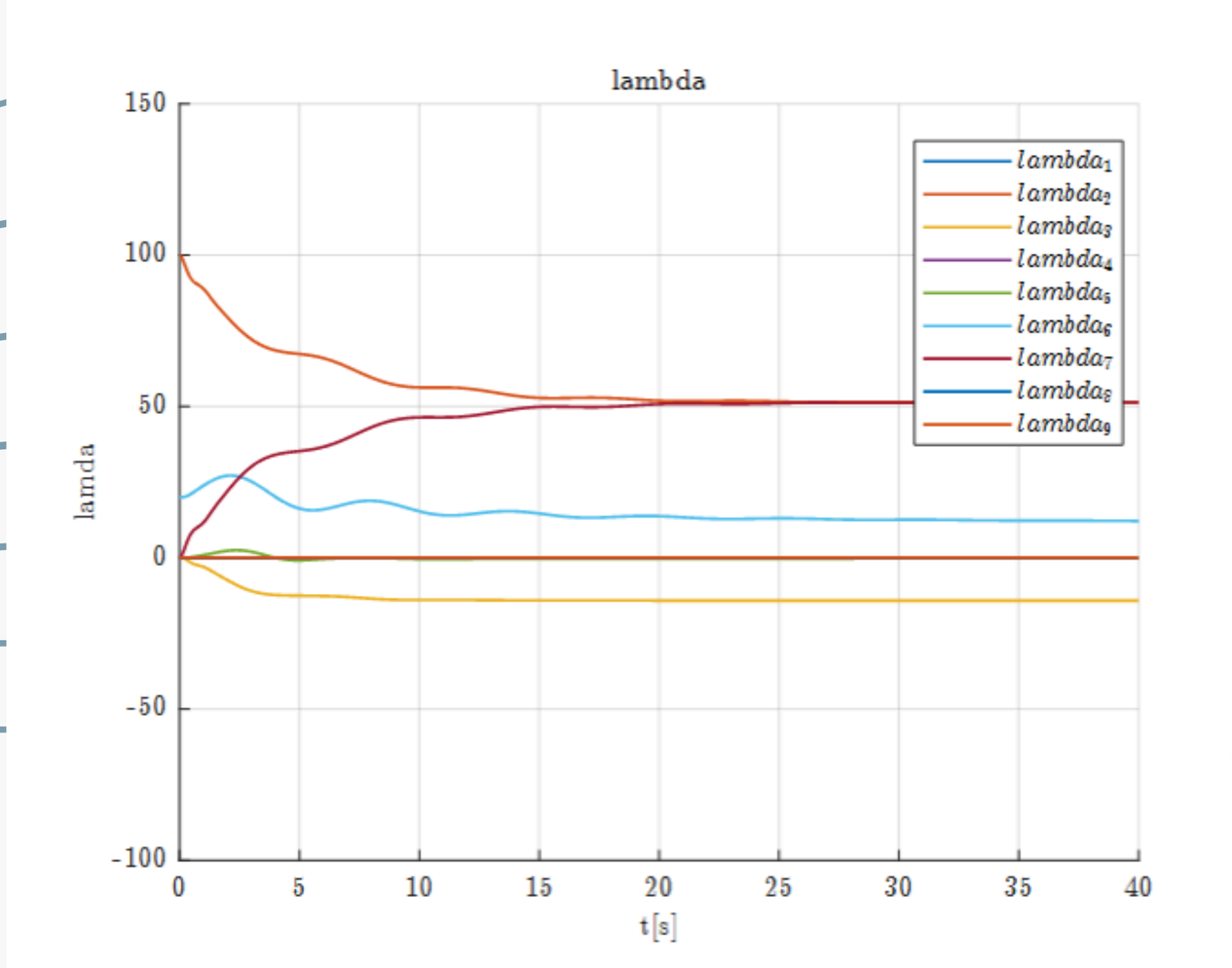
Regulation



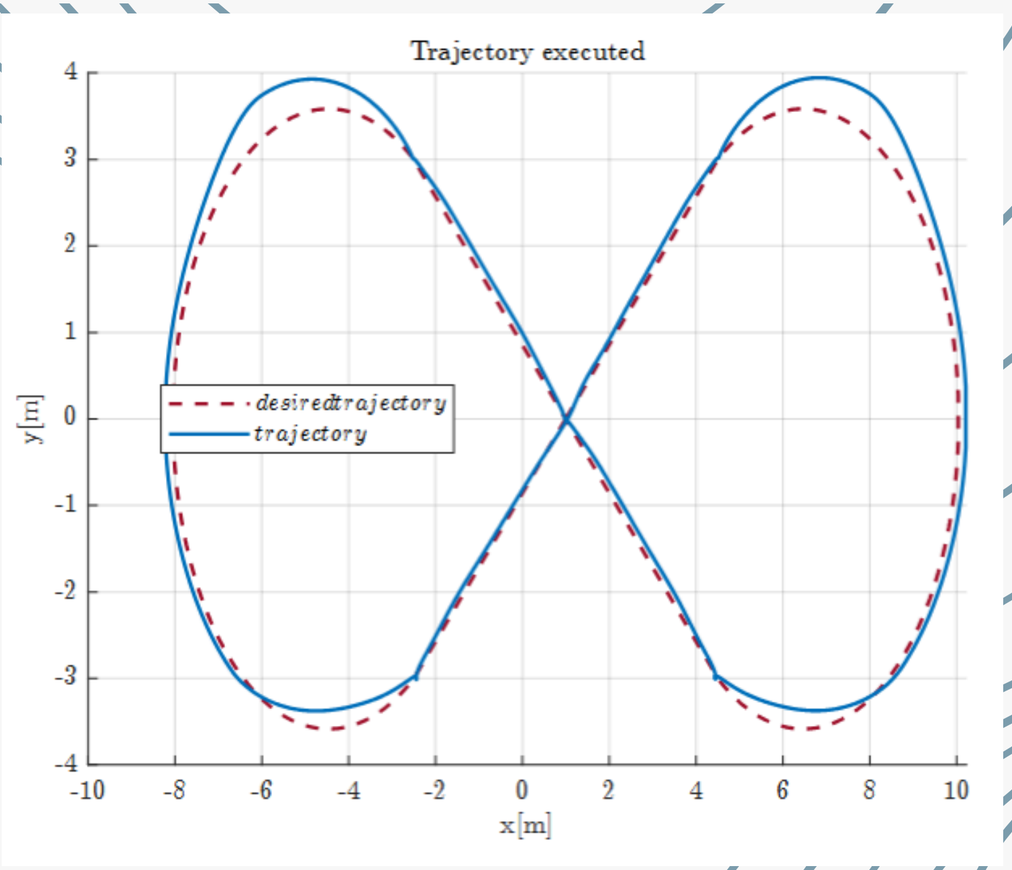


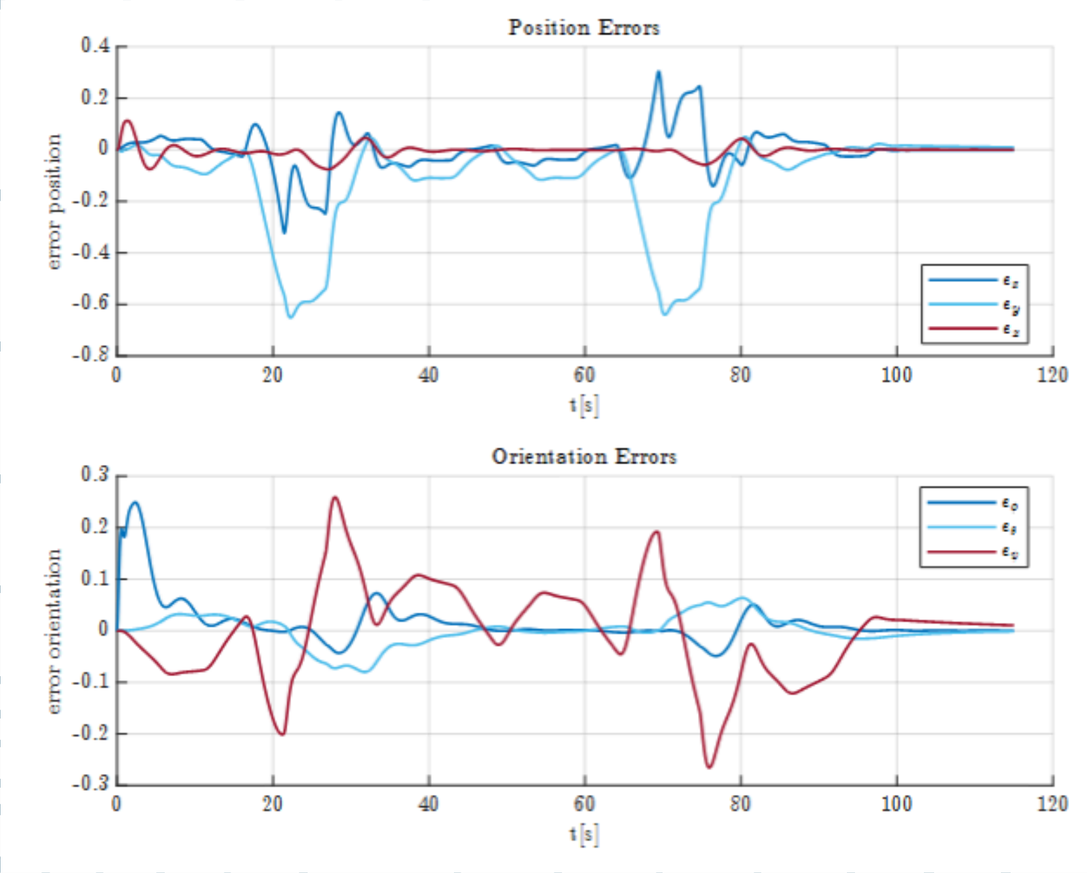
Adaptive Controller

Regulation



Tracking

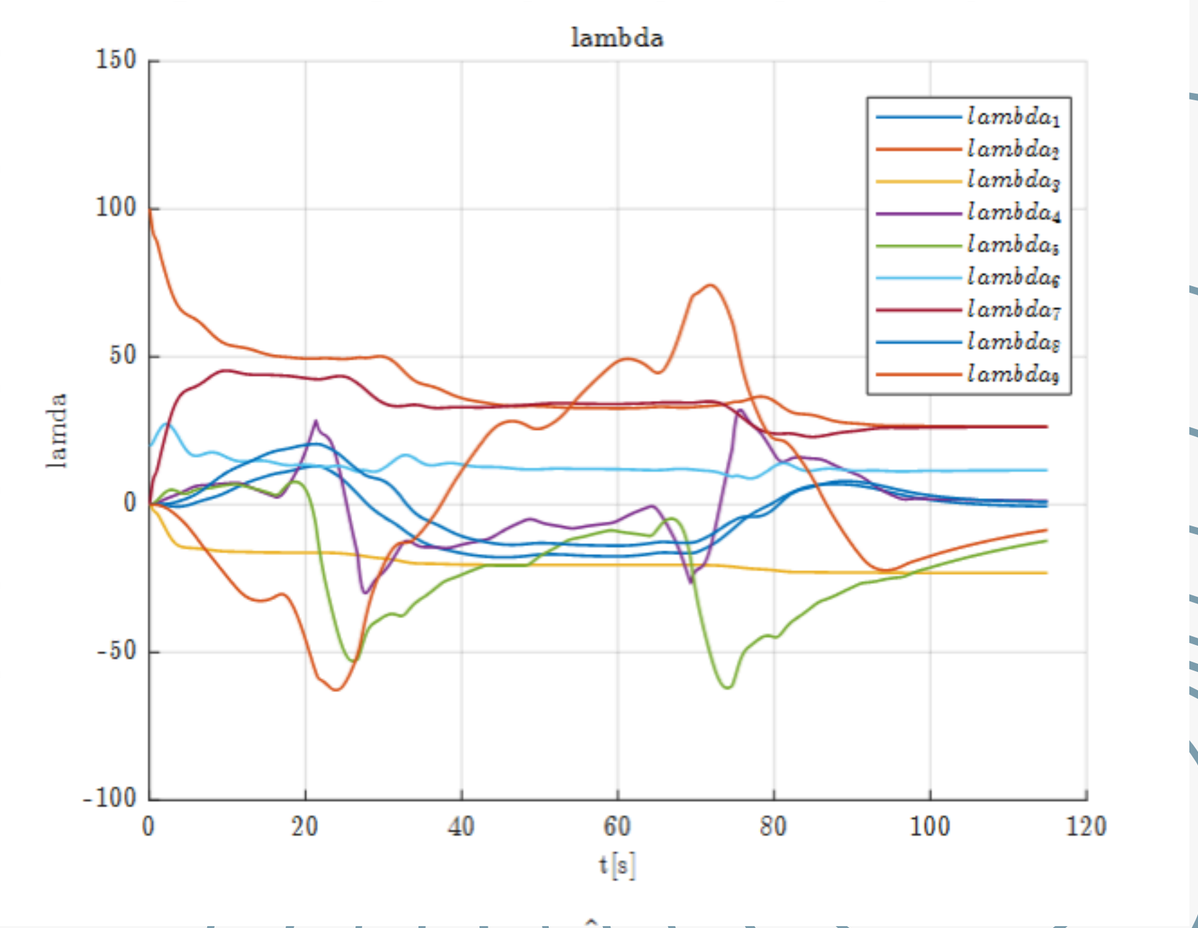




Adaptive Controller

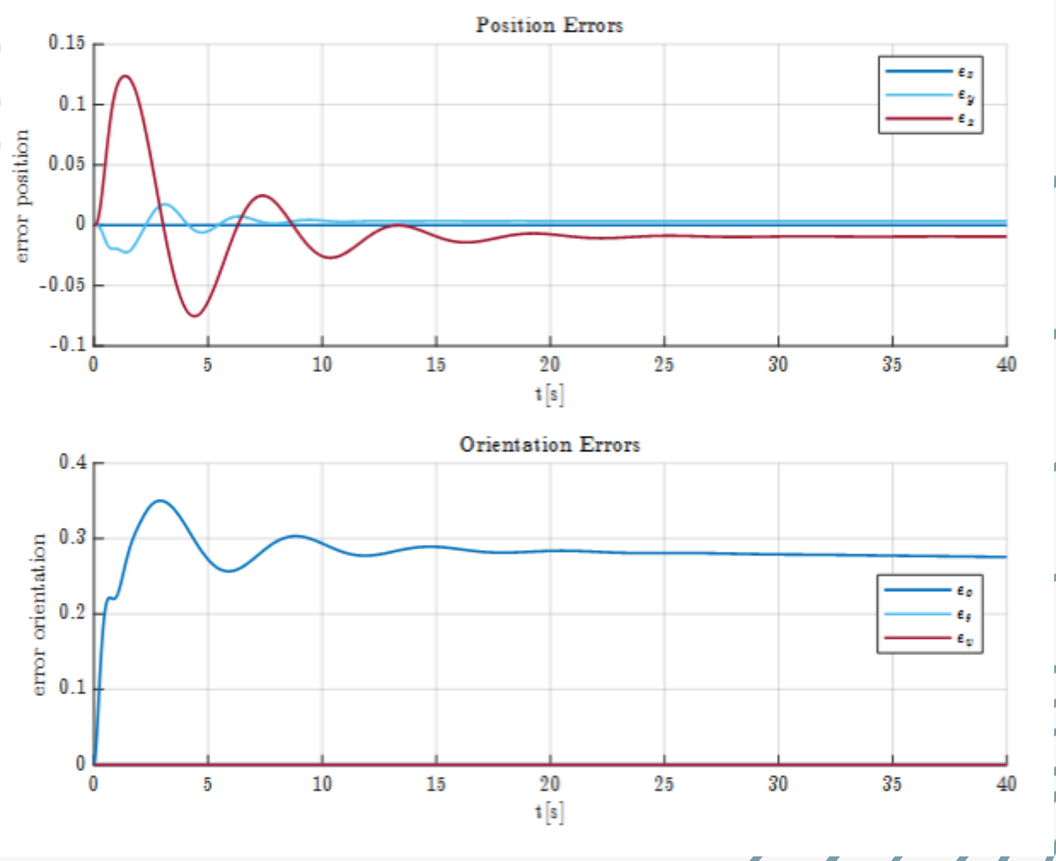
Adaptive Controller

Tracking

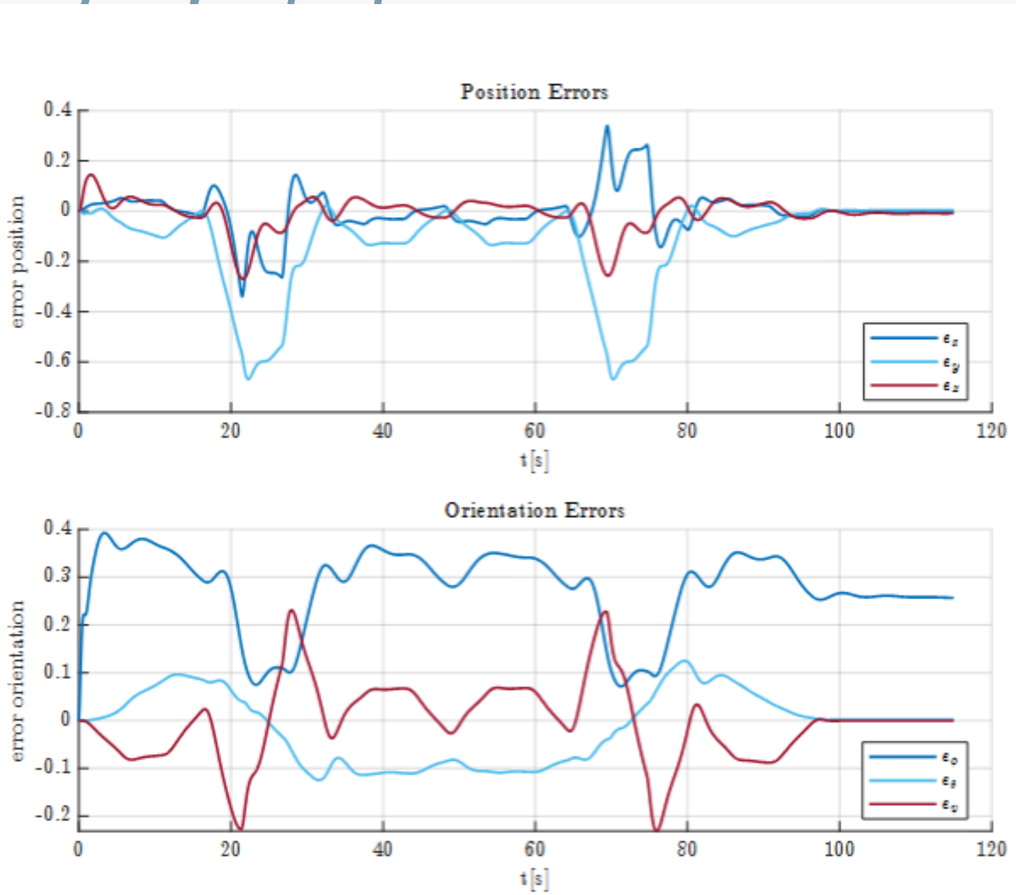


NOT so Adaptive Controller

Tracking



Regulation



Thanks

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